UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

# Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 015 · Applications of LDEs: simple harmonic oscillation · compiled 17 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S015·11 | Unit 2 deck 015 (“Applications of DE – SHM”), slide 11 (rendered .pptx) |
| T015 00:11:48 | Unit 2 AV-summary transcript “Application problems on linear differential equations – Simple harmonic oscillation” (16.3 min), segment start |
| CW U2 P37 | Unit 2 class-work problem 37 |

Contents

1. Part A · Formula sheet
2. 1 · The spring–mass model and simple harmonic motion
3. 2 · Solving spring problems
4. 3 · The damped oscillator
5. Part B · Symbol glossary

# A Formula sheet

## 1 The spring–mass model and simple harmonic motion

### E1.1 · Equilibrium condition

ESA ★★

$$mg = ks$$

GIVES $k = \frac{mg}{s} = \frac{W}{s}$ from a static stretch $s$.

APPLIES The mass hangs at rest.

UNITS $W$ in lb with $s$ in ft (lb/ft); $W$ in N with $s$ in m (N/m).

NOTE $W = mg$ is the weight.

S015·5 · S015·7 · S015·11 · T015 00:02:29

### E1.2 · Newton's second law for the hanging mass

ESA ★

$$mg - k(s+x) = m\frac{d^2x}{dt^2} \implies m\frac{d^2x}{dt^2} = -kx$$

GIVES The force balance; $mg$ and $ks$ cancel.

APPLIES $x$ measured from equilibrium, downward positive.

S015·6 · S015·7 · T015 00:04:30 · T015 00:05:03

### E1.3 · Equation of simple harmonic motion

ESA ★★

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0, \qquad \frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}$$

GIVES The model of the undamped, unforced spring.

APPLIES No damping, no external force.

NOTE A homogeneous LDE with constant coefficients: AE $m^2 + \omega^2 = 0$ (class 002, complex roots).

S015·7 · T015 00:05:15 · T015 00:05:33

### E1.4 · General solution

ESA ★★

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

GIVES The position at any instant.

APPLIES Roots $\pm i\omega$.

NOTE $c_1 = x(0)$, $c_2 = \frac{\dot{x}(0)}{\omega}$.

S015·8 · T015 00:06:28 · T015 00:06:46

### E1.5 · Amplitude–phase form

ESA ★★

$$x(t) = A\cos(\omega t + \varphi), \quad c_1 = A\cos\varphi,\ c_2 = -A\sin\varphi, \quad A = \sqrt{c_1^2 + c_2^2},\ \tan\varphi = -\frac{c_2}{c_1}$$

GIVES Amplitude and phase from $c_1$, $c_2$.

APPLIES After the constants are known.

NOTE Pick the quadrant of $\varphi$ from the signs of $\cos\varphi = \frac{c_1}{A}$ and $\sin\varphi = -\frac{c_2}{A}$.

S015·8 · T015 00:07:05

### E1.6 · Period and natural frequency

ESA ★★

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

GIVES Seconds per oscillation; oscillations per second.

APPLIES Undamped motion.

NOTE The lecturer: “you should remember all these formulae”. The last form of $f$ is added for clarity.

S015·9 · T015 00:07:40 · T015 00:07:58

### E1.7 · Maximum velocity

ESA ★★

$$\dot{x} = -A\omega\sin(\omega t + \varphi), \qquad v_{\max} = A\omega$$

GIVES The greatest speed, reached at the equilibrium position.

APPLIES Any SHM.

NOTE Maximum acceleration is $A\omega^2$ (added for clarity).

S015·22 · CW U2 P37

## 2 Solving spring problems

### E2.1 · Mass from weight

ESA ★★

$$m = \frac{W}{g}, \qquad g = 32\ \text{ft/s}^2 \text{ or } 9.8\ \text{m/s}^2$$

GIVES Mass in slugs (from lb) or kg.

APPLIES The weight is given in pounds or kilogram-weight.

NOTE A 12 lb weight has $m = \frac{12}{32} = \frac{3}{8}$ slug; a body of 10 kg has $m = 10$ kg.

S015·11 · S015·18 · T015 00:10:05

### E2.2 · Constants from the initial conditions

ESA ★★

$$c_1 = x(0), \qquad c_2 = \frac{\dot{x}(0)}{\omega}$$

GIVES The particular motion.

APPLIES Downward positive: pulled down $d$ → $x(0) = +d$; raised $d$ → $x(0) = -d$; upward velocity $v$ → $\dot{x}(0) = -v$; released → $\dot{x}(0) = 0$.

S015·13 · S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24

# 3 The damped oscillator

### E3.1 · Damped oscillator ESA ★

$$m\ddot{x} + 2\lambda mn\dot{x} + mn^2 x = 0 \implies \ddot{x} + 2\lambda n\dot{x} + n^2 x = 0$$

GIVES The model with a resistance $2\lambda mn\dot{x}$.

APPLIES Restoring force $mn^2x$ (so $n = \sqrt{k/m}$).

S015·15

### E3.2 · Roots and damped frequency ESA ★

$$r = -\lambda n \pm in\sqrt{1-\lambda^2}, \qquad \omega_d = n\sqrt{1-\lambda^2}$$

GIVES The underdamped CF $e^{-\lambda nt}(A\cos\omega_d t + B\sin\omega_d t)$.

APPLIES $0 < \lambda < 1$.

NOTE Slide 015·16 calls $\omega_d$ simply $\omega$.

S015·16

### E3.3 · Damped motion from $x(0)=0$, $\dot{x}(0)=x_0$ ESA ★

$$x(t) = \frac{x_0}{n\sqrt{1-\lambda^2}}\, e^{-\lambda nt} \sin\!\left(n\sqrt{1-\lambda^2}\, t\right)$$

GIVES The slide's boxed answer.

APPLIES $0 < \lambda < 1$.

NOTE Checked by substitution and both initial conditions (sympy).

S015·16

# B Symbol glossary

| Symbol | Name | Meaning | Unit | Appears in |
|---|---|---|---|---|
| $m$ | Mass | $W/g$; slug (ft–lb) or kg. Not the AE variable here (the AE uses $r$) | slug, kg | E1.3, E2.1 |
| $W$ | Weight | $mg$; pounds or kilogram-weight | lb, N | E2.1 |
| $g$ | Acceleration due to gravity | 32 ft/s² or 9.8 m/s² | ft/s², m/s² | E1.1, E2.1 |
| $k$ | Spring constant | Force per unit stretch | lb/ft, N/m | E1.1, E1.3 |
| $s$ | Static stretch | Extension at equilibrium, $mg = ks$ | ft, m | E1.1 |
| $x(t)$ | Displacement | From equilibrium, downward positive | ft, m | E1.3, E1.4 |
| $\dot{x}$, $\ddot{x}$ | Velocity, acceleration | $\frac{dx}{dt}$, $\frac{d^2x}{dt^2}$ | | E1.2, E2.2 |
| $\omega$ | Angular frequency | $\sqrt{k/m}$; on slide 015·16 the damped frequency | rad/s | E1.3, E3.2 |
| $A$ | Amplitude | $\sqrt{c_1^2 + c_2^2}$; on slide 015·16 an integration constant | ft, m | E1.5 |
| $\varphi$ | Phase | $\tan\varphi = -\frac{c_2}{c_1}$ | rad | E1.5 |
| $T$, $f$ | Period, frequency | $\frac{2\pi}{\omega}$, $\frac{1}{T}$ | s, Hz | E1.6 |
| $c_1$, $c_2$ | Arbitrary constants | $x(0)$, $\dot{x}(0)/\omega$ | | E1.4, E2.2 |
| $n$ | Undamped frequency | In $mn^2x$ (slide 015·15) | rad/s | E3.1 |
| $\lambda$ | Damping ratio | $0 < \lambda < 1$ on the slide | | E3.1, E3.2 |
| $x_0$ | Initial velocity | $\dot{x}(0)$ on slide 015·15 (not a position) | ft/s | E3.3 |

***Remark.*** The deck uses $x$, $\frac{dx}{dt}$, $\frac{d^2x}{dt^2}$ (slides 5–14) and $\dot{x}$, $\ddot{x}$ (slides 15–22); they mean the same. $s$ is the static stretch at equilibrium, $x$ the displacement from equilibrium. $\omega$ is the angular frequency $\sqrt{k/m}$ in the undamped case; on slide 015·16 the same letter is the damped frequency $n\sqrt{1-\lambda^2}$, and slide 015·15 calls the undamped one $n$. $A$ is the amplitude on slides 8–9 and 22 but an integration constant on slide 015·16. $c_1$, $c_2$ and $C_1$, $C_2$ are both arbitrary constants. $f$ is the frequency, not a function.